Ex. 12 End to end communication

2/10/24 at transport layer

Aim: To implement echo client server using TCP/UDP Sockets.

Code:

```
import Socket
import threading

def start_Server():
    server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    server.bind(('127.0.0.1', 5000))
    server.listen(1)
    print("[Server] Listening on 127.0.0.1:5000")
    conn, addr = server.accept()
    print(f"[Server] Connected by {addr}")
    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower() == 'quit':
            print("[Server] Connection closed by client.")
            break
        print(f"[Server] Received: {data}")
        conn.send(f"Server received: {data}".encode())
    conn.close()
```

```
def start_client():
    client = socket.socket(socket.AF_INET, socket.SOCK_
                                        STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] Connected to server")

while True:
    msg = input("Enter message (or 'quit' to
                                        exit): ")
    client.send(msg.encode())
    if msg.lower() == 'quit':
        break
    response = client.recv(1024).decode()
    print(f"[SERVER RESPONSE] {response}")

client.close()
if __name__ == "__main__":
    server_thread = threading.Thread(target=
        start_server, daemon=True)
    server_thread.start()
    start_client()
```

Output:

[SERVER] Listening on 127.0.0.1:5000 [CLIENT] connected to server.

[SERVER] connected by ('127.0.0.1', 53458) Enter message (or 'quit' to exit):

hello

[SERVER] Received: hello

[SERVER RESPONSE] Server Received: hello

Enter message (or 'quit' to exit):

[SERVER] Listening on 127.0.0.1:5000 [CLIENT] connected to server

[SERVER] connected by ('127.0.0.1', 60328) Enter message (or 'quit' to exit):

HI

[SERVER] Received: Hi

[SERVER RESPONSE] Server received: HI

Enter message (or 'quit' to exit): quit

[SERVER] connection closed by client

UDP

```
import socket
import threading

def udp_server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] client {addr} ended the connection")
            break
        print(f"[SERVER] Received from {addr}: {msg}")
        s.sendto(data, addr)
    s.close()
    print("[SERVER] Stopped.")

def udp_client():
    c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
```

```
Socket. SOCK_ DGRAM)
Server = ("127.0.0.1", 12000)
while True:
    msg = input("Enter message ('out' to
                                    stop):")
    c.sendto(msg.encode(), server)
    if msg.lower() == 'out':
        print("Client closing connection...")
                                    break
    data, _ = c.recvfrom(1024)
    print("Echo from Server:", data.decode()
    c.close())
print("CLIENT STOPPED.")
if __name__ == "__main__":
    threading.Thread(target = udp_server,
                     daemon=True).start()
                                    udp_client()
```

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op:

Enter message (or 'out' to stop): [server] Running on 127.0.0.1:12000 hi

[server] Received from ('127.0.0.1', 49670): hi

Echo from Server: hi

Enter message (or 'out' to stop) hello

[Server]: Received from ('127.0.0.1', 49670): hello

Echo from server: hello

Enter message (or 'quit' to stop): how are you

[Server] Received from ['127.0.0.01' 49670): how are you

Echo from Server: how are you.

Enter message (or 'quit' to stop): quit

[client] closing connection ... [server] client ['127.0.0.1', 49670) ended the connection.

[client] stopped. [Server] stopped.

Result:

Thus the end - to end communication was implemented.